CHAPTER 2

PLANT KINGDOM

CHAPTER ESSENCE AT A GLANCE:
Evolutionary transition from thalloid non-vascular bryophytes and pteridophytes to seed-bearing vascular gymnosperms and angiosperms.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Algal Pigmentation and Food Reserves:** Algae are classified into three major classes by photosynthetic pigments and storage polysaccharides: Chlorophyceae (Chlorophyll a and b, starch storage, cellulose cell wall), Phaeophyceae (Chlorophyll a and c with fucoxanthin brown pigment, laminarin or mannitol storage, algin outer wall), and Rhodophyceae (Chlorophyll a and d with r-phycoerythrin red pigment, floridean starch storage).

• **Bryophyte Gametophytic Dominance:** Bryophytes are the amphibians of the plant kingdom requiring water for flagellated antherozoids to fertilize non-motile archegonial eggs. The haploid gametophyte represents the independent dominant photosynthetic phase, upon which the diploid sporophyte remains nutritionally dependent throughout its lifespan.

• **Pteridophyte Vascularization and Heterospory:** Pteridophytes represent the first terrestrial vascular plants with xylem and phloem. While homosporous ferns produce identical spores, heterosporous genera like Selaginella and Salvinia produce distinct microspores and megaspores, retaining female gametophytes on parent sporophytes as an evolutionary precursor to the seed habit.

• **Gymnosperm Naked Seeds:** Gymnosperms possess exposed ovules unshielded by ovary walls both before and after fertilization, yielding naked seeds. Roots exhibit mutualistic fungal mycorrhizae in Pinus and nitrogen-fixing cyanobacterial coralloid roots in Cycas. Microspores develop inside microsporangiate pollen cones.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Algal Classes} = \text{Chlorophyceae (Starch)} + \text{Phaeophyceae (Mannitol)} + \text{Rhodophyceae (Floridean)}$$

Comparative Pigmentation and Stored Food Reserves

$$\text{Seed Habit Precursor} = \text{Heterospory} \rightarrow \text{Microspores} + \text{Megaspores on Parent Sporophyte}$$

Evolution of Seed Habit in Heterosporous Pteridophytes

$$\text{Bryophyte Life Cycle} = \text{Gametophyte (Haploid Dominant)} + \text{Sporophyte (Parasitic Dependent)}$$

Haplodiplontic Phase Alternation in Bryophytes

$$\text{Gymnosperm Cones} = \frac{\text{Microsporophylls (Pollen Cones)}}{\text{Megasporophylls (Ovuliferous Seed Cones)}}$$

Unisexual Strobilus Dimorphism in Conifers

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never state that gymnosperms bear fruits; gymnosperms completely lack an ovary wall, so their fertilized ovules form naked seeds.

• **Alert:** Remember that the dominant free-living generation is the haploid gametophyte in bryophytes, but the diploid sporophyte in pteridophytes and gymnosperms.

• **Alert:** Heterospory in Selaginella and Salvinia is an essential evolutionary step toward the seed habit because female gametophytes develop within the parent sporophyte.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Why are bryophytes commonly referred to as the amphibians of the plant kingdom? Name the asexual reproductive buds found in Marchantia. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Bryophytes live in soil but depend fundamentally on an external film of water for flagellated male antherozoids to swim to the archegonium for sexual fertilization. **[1 Mark]**

• The multicellular green asexual reproductive buds found in Marchantia are termed gemmae (produced inside gemma cups). **[1 Mark]**

2. Compare Chlorophyceae, Phaeophyceae, and Rhodophyceae based on major photosynthetic pigments and stored food reserves. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Chlorophyceae: Chlorophyll a and b pigments; food stored as starch. **[0.75 Mark]**

• Phaeophyceae: Chlorophyll a and c with fucoxanthin; food stored as complex carbohydrates laminarin or mannitol. **[0.75 Mark]**

• Rhodophyceae: Chlorophyll a and d with r-phycoerythrin; food stored as floridean starch structurally similar to amylopectin and glycogen. **[0.5 Mark]**

3. Explain heterospory with reference to Selaginella. Why is heterospory considered an important evolutionary precursor to the seed habit? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Heterospory is the production of two distinct kinds of spores by a plant: small microspores that germinate into male gametophytes, and large megaspores that germinate into female gametophytes. **[1 Mark]**

• In Selaginella and Salvinia, the megaspore is retained within the megasporangium on the parent sporophyte during female gametophyte development. **[1 Mark]**

• The development of the zygote into a young embryo takes place within the female gametophyte on the parent sporophyte, establishing the foundation of seed evolution. **[1 Mark]**

4. Distinguish between the root adaptations of Cycas and Pinus with respect to their microbial symbiotic associations. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Cycas roots: Possess specialized apogeotropic coralloid roots that establish symbiotic associations with nitrogen-fixing cyanobacteria such as Anabaena and Nostoc. **[1.5 Marks]**

• Pinus roots: Form obligate symbiotic fungal associations termed mycorrhizae that enhance phosphorus and water absorption from the soil; Pinus seeds cannot germinate without mycorrhizal association. **[1.5 Marks]**

5. Tabulate three distinct differences between the gametophyte of a Bryophyte and the gametophyte of a Pteridophyte. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Dominance: Bryophyte gametophyte is the dominant, independent, photosynthetic phase; Pteridophyte gametophyte (prothallus) is short-lived and subordinate to the dominant sporophyte. **[1 Mark]**

• Vascular tissue: Both gametophytes lack vascular tissues, but the bryophyte gametophyte supports the dependent sporophyte directly on its body. **[1 Mark]**

• Differentiation: Bryophyte gametophyte is differentiated into thalloid or leafy shoots with rhizoids; Pteridophyte gametophyte is an inconspicuous, thalloid, heart-shaped prothallus. **[1 Mark]**

6. Match the following economically valuable botanical products with their respective plant groups: (a) Algin, (b) Carrageen, (c) Peat, (d) Agar, (e) Coralloid roots, (f) Microscopic unicellular algae. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• (a) Algin: Brown algae (Phaeophyceae, e.g., Laminaria, Fucus). **[0.5 Mark]**

• (b) Carrageen: Red algae (Rhodophyceae, e.g., Chondrus). **[0.5 Mark]**

• (c) Peat: Bryophytes (Sphagnum moss). **[0.5 Mark]**

• (d) Agar: Red algae (Gelidium and Gracilaria). **[0.5 Mark]**

• (e) Coralloid roots: Gymnosperms (Cycas). **[0.5 Mark]**

• (f) Chlorella or Chlamydomonas: Green algae (Chlorophyceae). **[0.5 Mark]**

7. Contrast Gymnosperms with Angiosperms regarding: (a) Ovule protection, (b) Xylem conducting elements, (c) Phloem transport cells, and (d) Endosperm ploidy. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• (a) Ovule protection: Gymnosperm ovules are naked and freely exposed on megasporophylls; Angiosperm ovules are enclosed within the protective ovary wall. **[1 Mark]**

• (b) Xylem conducting elements: Gymnosperm xylem consists predominantly of tracheids lacking true xylem vessels (except in Gnetales); Angiosperm xylem contains efficient vessel elements. **[1 Mark]**

• (c) Phloem transport cells: Gymnosperm phloem possesses albuminous cells and sieve cells lacking true companion cells; Angiosperm phloem possesses companion cells and sieve tube elements. **[1 Mark]**

• (d) Endosperm ploidy: Gymnosperm endosperm is formed before fertilization and is haploid (n); Angiosperm endosperm is formed following double fertilization and triple fusion and is triploid (3n). **[1 Mark]**